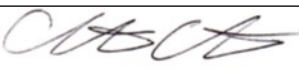


Guideline Title: Pediatric Cyanotic Congenital Heart Disease

Guideline Number: GUID-3203-PED007

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

I. **PURPOSE:** Restoration of cardiac output to improve tissue oxygenation and inadequate perfusion by maintaining an open ductus arteriosus and expanding intravascular volume as appropriate.

II. **PATHOPHYSIOLOGY:**

Certain congenital heart disease conditions exist resulting in circulatory shunts between the right heart and left heart bypassing the lungs. Cyanosis ensues that is not responsive to application of supplemental oxygenation.

After birth, in response to lung expansion and gas exchange, the ductus arteriosus gradually closes over the first one to two weeks of life. While the ductus arteriosus remains open, unoxygenated blood being shunted from right to left travels "backward" from the aorta into the pulmonary artery providing for lung perfusion and gas exchange. As the ductus closes, this circulatory pattern ceases, and the infant with congenital shunting cannot oxygenate blood. Maintaining an open ductus arteriosus can be lifesaving until the congenital abnormality can be corrected surgically.

These ductal-dependent cyanotic congenital heart disease conditions include: coarctation of the aorta, transposition of the great vessels, tricuspid atresia, tetralogy of Fallot, and total anomalous pulmonary venous return.

III. **DEFINITIONS:**

IV. **DEPARTMENT GUIDELINE:**

- A. Assess airway, breathing, and circulation as per *GUID3203-PED001 – Pediatric General Management*.
- B. Apply cardiac monitor, monitor pulse, blood pressure, and oximetry. Assess oxygenation with pre-ductal and post-ductal pulse oximetry readings on the upper and lower extremities, respectively.
- C. Apply oxygen for hypoxia and determine response. If the response is worsening perfusion, remove the oxygen. Avoid hyperventilation and hyperoxia.
- D. Monitor ETCO₂ if intubated.
- E. Obtain the largest IV access for the patient's size, two if possible. If unable to obtain IV access, obtain IO access per *GUID3203-PR013 – Intraosseous Infusions – EZIO*.
- F. If no improvement in hypoxia with oxygen, consider discussion for orders from the sending or receiving physician for prostaglandin E₁, alprostadil, infusion. Carefully monitor patient for hypotension and potential intubation/airway management due to side effects of Prostaglandin administration.

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- i. **prostaglandin E₁ (alprostadil) infusion at 0.1 mcg/kg/min**, and titrate to improved oxygenation and systemic perfusion, max dose 0.4mcg/kg/min.
 - G. If signs of pulmonary vascular congestion and/or fluid overload are present, consult with physician to withhold fluid bolus and administer **Furosemide 1 mg/kg IV**. Maximum single dose 20mg IV.
 - H. If no evidence of fluid overload:
 - i. Resuscitate with 20 ml/kg IV crystalloid bolus over 5-10 minutes. Repeat up to two times for a total of 60 ml/kg if patient remains in shock unless signs of fluid overload are present.
 - ii. For patients failing initial fluid bolus, initiate inotropic support:
 - 1. **Dopamine infusion at 5 - 20 mcg/kg/min IV/IO**
 - 2. **Dobutamine infusion at 2 - 20 mcg/kg/min IV/IO**
 - 3. **Milrinone infusion at 0.25 - 0.75 mcg/kg/min IV/IO**
 - 4. **Epinephrine infusion at 0.05 - 2 mcg/kg/min IV/IO**
 - I. If the blood glucose level is less than 60, treat per *GUID3203-M021 - Hypoglycemia*.
 - J. Monitor urine output by indwelling urinary catheter if available. Titrate resuscitation to 1 ml/kg/hr.
- V. **DOCUMENTATION:**
 - A. EMS Charts
- VI. **REFERENCES:**
 - A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.